

THE POPULATION FREQUENCIES OF SPECIES AND THE ESTIMATION OF POPULATION PARAMETERS

By I. J. GOOD

A random sample is drawn from a population of animals of various species. (The theory may also be applied to studies of literary vocabulary, for example.) If a particular species is represented r times in the sample of size N , then r/N is not a good estimate of the population frequency, p , which is the probability that a particular species will be represented r times in a sample of size N . Methods are given for estimating p , assuming virtually nothing about the underlying population. The estimates are expressed in terms of smoothed values of the numbers n_r ($r = 1, 2, \dots$), where n_r is the number of distinct species that are each represented r times in the sample. (For example, if $n_1 = 10$, as 'the frequency of the frequency r '.) Turing is acknowledged for the most interesting part of the work. An estimate of the proportion of the population represented by the species in the sample is an immediate corollary. Estimates are made of measures of the population, including Yule's 'characteristic' and Shannon's 'entropy'. Methods are given for testing hypotheses about the underlying population. It is here that the theory does depend on assumptions about the underlying population. It is here that the theory is applied by other writers. It is pointed out that a hypothesis can give a good fit to the data, but quite the wrong value for Yule's characteristic. An example of this is Fisher's analysis of Williams's on *Macrolepidoptera*.

1. Introduction. We imagine a random sample to be drawn from a population of animals of various species. Let the sample size be N and let n_r be the number of species represented exactly r times in the sample, so that

$$\sum_{r=1}^{\infty} r n_r = N.$$

The sample tells us the values of $n_1, n_2, \dots$, but not of n_0 . In fact it is not possible to estimate n_0 from the sample, but n_0 should be finite though we shall find it convenient to suppose that it is



THE NUMBER OF NEW SPECIES, AND THE INCREASE IN POPULATION COVERAGE, WHEN A SAMPLE IS INCREASED

By I. J. GOOD AND G. H. TOULMIN

A sample of size N is drawn at random from a population of animals of various species. Methods are given for estimating, knowing only the contents of this sample, the number of species which will be represented r times in a second sample of size λN ; these also enable us to estimate the number of different species and the proportion of the whole population represented in the second sample. A formula is found for the variance of the estimate; when $\lambda = 2$, this variance becomes in general very large, so that the estimate is useless without modification. This difficulty can be partly overcome, at least for $\lambda < 5$, by using Euler's method with a suitable parameter or the methods described by Shanks (1955) to hasten the convergence of the series by which the estimate is expressed. The methods are applied to the words from *Our Mutual Friend*, to an entomological sample, and to a sample of Macaulay's essay on Bacon.

1. INTRODUCTION

This is a further development of the theory expounded by Good (1953); that paper will be referred to, for brevity, by the letter G throughout. In the present paper a random sample of size N , the *basic sample*, to be drawn from an infinite population of animals of various species, and suppose that n_r distinct species are each represented exactly r times in the sample, so that

$$\sum_{r=1}^{\infty} r n_r = N. \tag{1}$$
$$d = \sum_{r=1}^{\infty} n_r,$$

Notion of **sample coverage**

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Prediction of species on a further sample

Fisher, Corbet
and Williams
(1943)

Good
(1953)

Good and
Toulmin
(1956)

Fisher's log-
series model

Prediction of N.
of species in a
further sample